

PONMUDI M R

Electronics & Communication Engineering Student | Embedded Systems & RF/SDR Engineer

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PROFESSIONAL SUMMARY

Electronics and Communication Engineering student with hands-on experience in embedded systems, RF/SDR signal processing, IoT, and PCB design. Built and led award-winning hardware projects spanning LoRa disaster communication, GPS-denied satellite positioning, and custom PCB fabrication — taken from prototype to national-level competition and government research submission (DRDO/iDEX, IEEE, Ministry of Education).

EDUCATION

B.E. Electronics and Communication Engineering

Bannari Amman Institute of Technology (BIT), Sathyamangalam, Tamil Nadu | 2026 – 2029

TECHNICAL SKILLS

- **Programming:** C, Python, Arduino Programming, Embedded C (learning)
- **Embedded Systems:** ESP32, Arduino Mega, Raspberry Pi, sensor interfacing
- **RF / Communication:** LoRa, UART, SPI, I²C, Wi-Fi, Software-Defined Radio (RTL-SDR, SDR++), Doppler-shift / RF signal analysis
- **PCB Design:** KiCad, schematic design, PCB layout, fabrication, soldering
- **Machine Learning:** YOLOv8-based signal classification for RF datasets
- **Operating Systems:** Fedora Linux, Ubuntu Linux
- **Tools:** Git, GitHub, VS Code, Arduino IDE, Google Apps Script

PROJECTS

TribalNet — RF-Based Disaster Communication System | *ESP32, LoRa SX1278 (433 MHz), Embedded Systems*

- Designed a dual-node (RANGER/VILLAGE) LoRa mesh network for emergency communication in disaster-hit and remote areas, with MAX7219 LED matrix alerts.
- Won 1st Prize (₹50,000) at NHIDE 2026; recognized as IEEE YESIST12 2026 National Finalist (Submission ID 3280); IPR filing initiated via BIT Sathy's IPR cell.
- Submitted to the YUKTI National Innovation Repository (Ministry of Education).

Cognitive Opportunistic LEO-PNT Receiver | *RTL-SDR, SDR++, Python, YOLOv8, RF Signal Processing*

- Built a GPS-denied positioning system using Doppler shift measurements from LEO satellites (Orbcomm, NOAA, Iridium).
- Developed a synthetic RF dataset generator and trained a YOLOv8 classifier to identify signal types (BPSK, QPSK, SDPSK, TACAN, and more).
- Submitted to iDEX/DRDO as part of Team 404's defense-tech research effort.

Auto Internet Login (nlogin) | *Linux, Bash, Git*

- Built a Linux CLI tool automating login/deauth for a Sophos/Cyberoam captive portal; published as an open-source project.

Police Siren PCB | *KiCad, PCB Design*

- Designed, fabricated, assembled, and tested a custom PCB, gaining full hands-on experience with the PCB development workflow.

Kanaku Book — Digital Ledger PWA | *Progressive Web App, Google Sheets / Apps Script*

- Built and published a payment-ledger PWA for a family textile business, using Google Sheets/Apps Script as the backend.

RC Aircraft | *Brushless Flight-Control Components*

- Built a brushless RC airplane using commercial flight-control components; handled electronic integration and system assembly.

ACHIEVEMENTS

- 2nd Prize — ChronoBIT'26 Hackathon, Team Zero Hour — built a 555 timer + CD4026 digital clock.
- Volunteer Coordinator — Smart India Hackathon (Hardware Edition).
- Active open-source contributor with multiple published GitHub projects.

LEADERSHIP & AFFILIATIONS

- Team Lead, Team 404 — leads a 3-member team (with Dinesh and Prateeksha) across hackathons and research submissions.
- Member, IEEE Electron Devices Society; AViNYA Club.
- Google Gemini Student Ambassador.

ADDITIONAL INFORMATION

Interests: Embedded Systems, IoT, RF Communication, Software-Defined Radio, LoRa Networks, PCB Design, Wireless Sensor Networks, Linux, Open-Source Development

Soft Skills: Problem Solving, Leadership, Team Collaboration, Technical Documentation, Project Management, Communication

Languages: English, Tamil